

Audio模块 8676 vs 8668

AFE HW 能力对比

MT8676 AFE HW Capability

- 下行**24ch**: 4组2ch I2S out, 1组8ch TDM out, 1组8ch Internal DP-TX out
- 上行**18ch**: 5组2ch I2S in, 1组8ch TDM in

In/Out	Max Valid Format	Mem IF (FE)	Audio IF (BE)
Max 24ch Output	2ch/192k/32b	DL_2ch_X	I2SOUT0: 2ch
	2ch/192k/32b	DL_2ch_X	I2SOUT1: 2ch
	2ch/192k/32b	DL_2ch_X	I2SOUT2: 2ch
	2ch/192k/32b	DL_2ch_X	I2SOUT6: 2ch
	8ch/48k/32b	DL_8CH_0 (8ch)	I2SOUT4: 8ch TDM
Max 18ch Input	8ch/96k/32b	DL_TDMOUT (8ch)	No PIN out, only for internal DP-TX
	2ch/192k/32b	VUL_2CH_X	FMI2S: 2ch master & slave
	2ch/192k/32b	VUL_2CH_X	I2SIN0: 2ch
	2ch/192k/32b	VUL_2CH_X	I2SIN1: 2ch
	2ch/192k/32b	VUL_2CH_X	I2SIN2: 2ch
	2ch/192k/32b	VUL_2CH_X	I2SIN6: 2ch
	8ch/48k/32b	VUL_CM0 (8ch)	I2SIN4: 8ch TDM

MT8668 AFE HW Capability

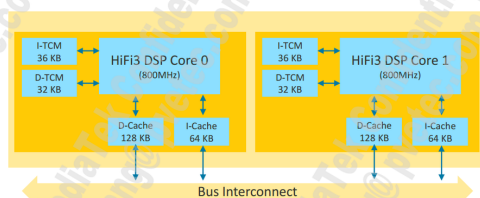
- 下行**36ch**: 2组2ch I2S out, 1组32ch TDM out
- 上行**48ch**: 4组2ch I2S in, 1组32ch TDM in, 1组8ch slave HS-I2S in

In/Out	Typical Usage	Mem IF (FE)	Audio IF (BE)
Max 36ch Output	2ch/48k/32b * 1 Data PIN	DL_2CH_X	I2SOUT1: 2ch
	2ch/48k/32b * 1 Data PIN	DL_2CH_X	I2SOUT2: 2ch
	8ch/48k/32b * 4 Data PIN	DL_48CH_0(32ch)	I2SOUT6: 32ch TDM (Master & Slave)
Max 48ch Input	2ch/48k/32b * 1 Data PIN	VUL_2CH_X	FMI2S: 2ch (Master & Slave)
	2ch/48k/32b * 1 Data PIN	VUL_2CH_X	I2SIN0: 2ch
	2ch/48k/32b * 1 Data PIN	VUL_2CH_X	I2SIN1: 2ch
	2ch/48k/32b * 1 Data PIN	VUL_2CH_X	I2SIN2: 2ch
	8ch/48k/32b * 4 Data PIN	VUL_CM2(32ch)	I2SIN6: 32ch TDM (Master & Slave)
	2ch/2048k/18b * 4 Data PIN	CQDMA	I2SIN_DMA0 (Slave only)

ADSP HW 能力对比

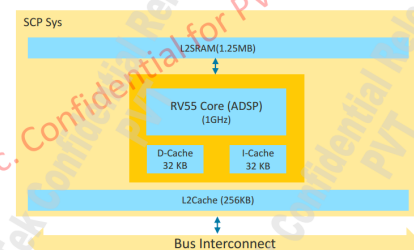
MT8676 ADSP HW IP

- HiFi3 Core 0 @800MHz
 - 64KB I-Cache
 - 128KB D-Cache
 - 36KB I-TCM
 - 32KB D-TCM
- HiFi3 Core 1 @800MHz
 - 64KB I-Cache
 - 128KB D-Cache
 - 36KB I-TCM
 - 32KB D-TCM
- Clock Rate: 800MHz

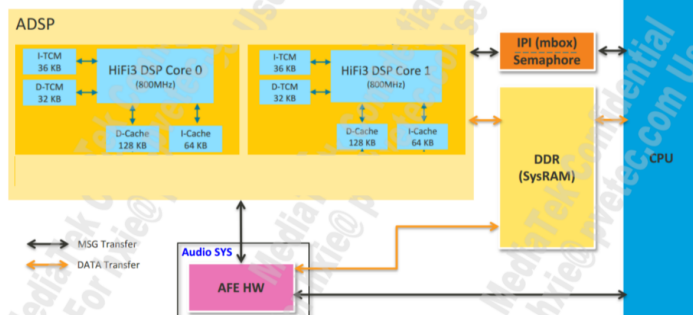


MT8668 ADSP HW IP Capability

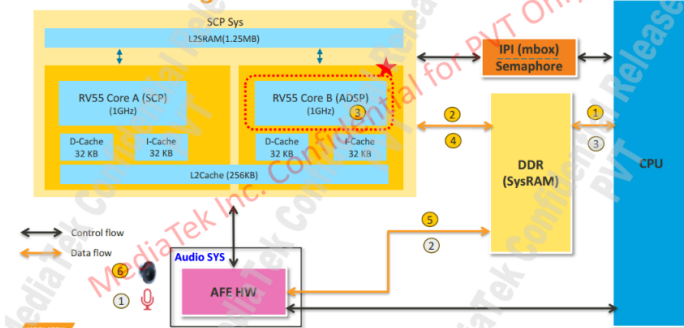
- RV55 ADSP on SCP Sys
 - RV55 Core @ 1GHz
- Cache
 - I-Cache 32 KB
 - D-Cache 32 KB
 - L2 Cache 256KB (shared)
- L2SRAM
 - 1.25MB (shared)



ADSP Block Diagram

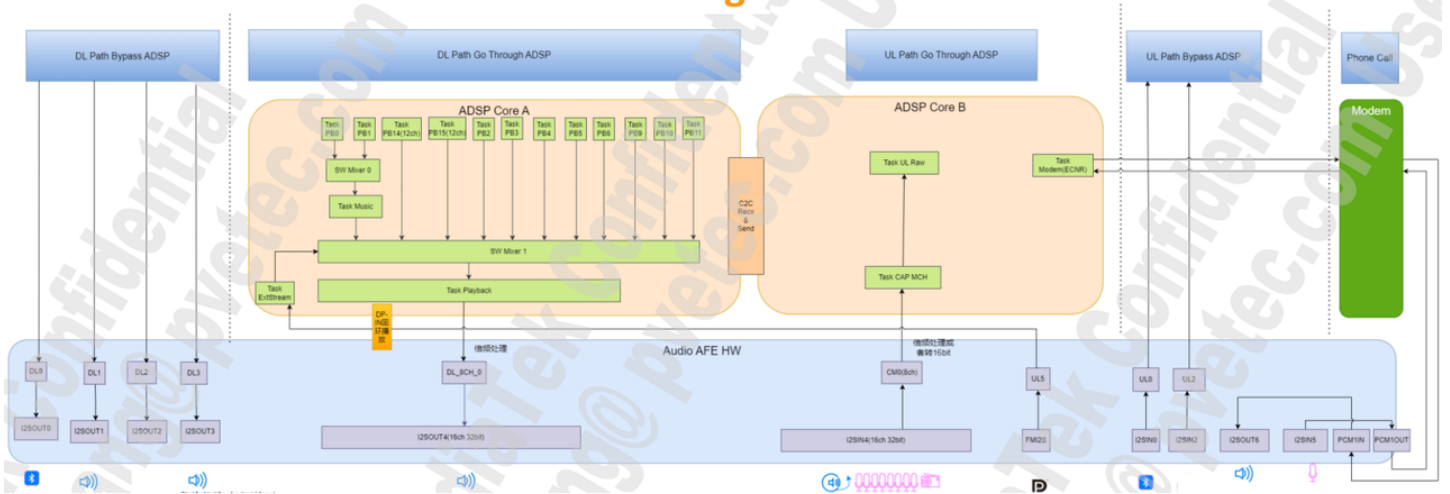


ADSP Block Diagram

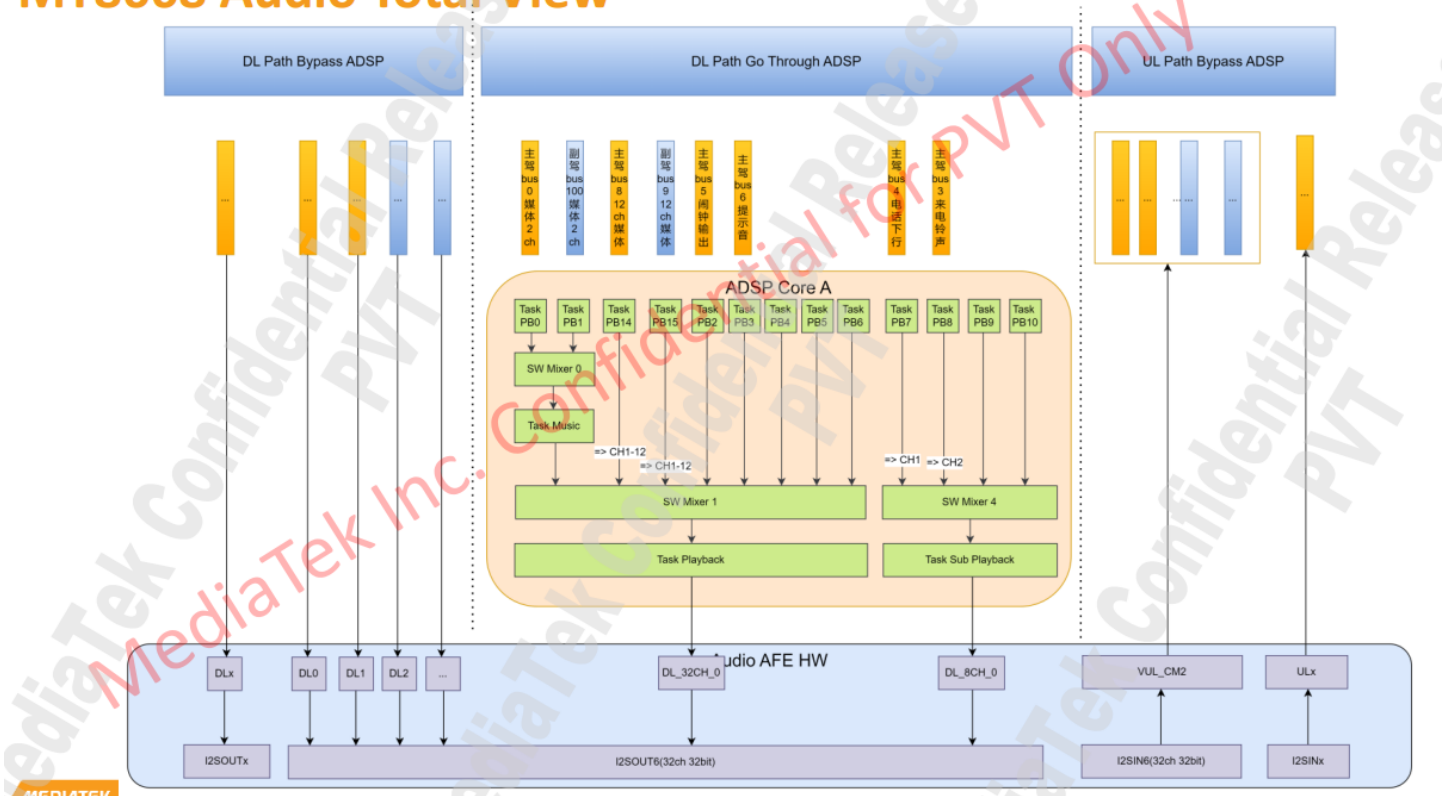


Audio通路参考设计

MT8676 Design



MT8668 Audio Total View



8676

1. ADSP: A核输出通道，不倍频可以输出8ch，倍频可以到16ch

B核输入通道接入CM0 UL，不倍频可以输入8ch，倍频可以到16ch

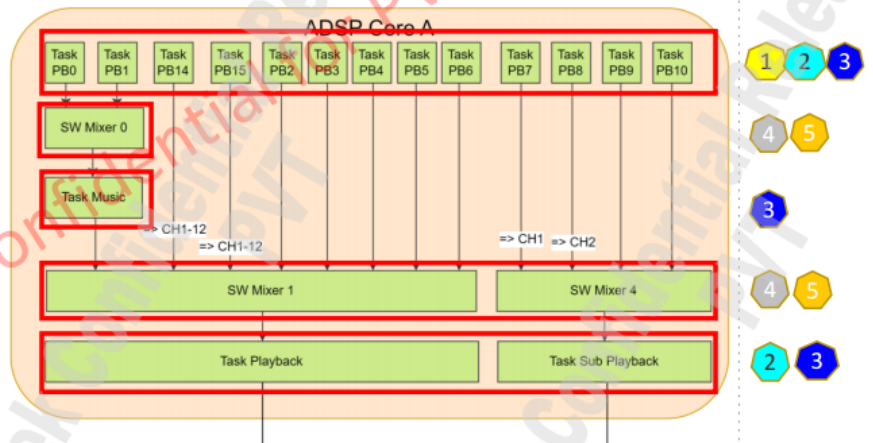
8668

1. 输出通道不需要倍频，即可输出32ch
2. 输入通道都不经过adsp，直接从AFE 给到hal。
3. 不具备在adsp Loopback功能

8676 vs 8668, audio_device.xml配置会有差异

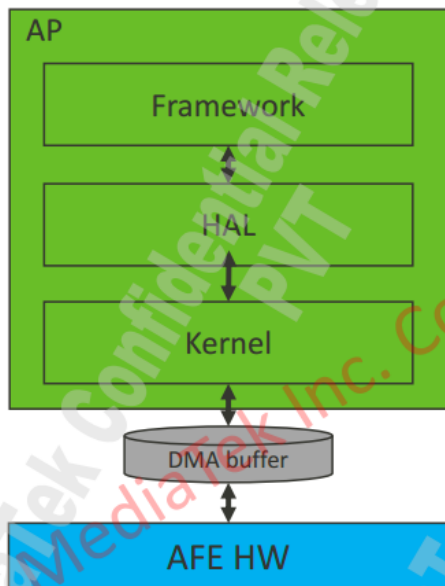
ADSP Internal SW View (DL)

1. DL Stream Tasks for bus
2. Volume process
3. Aurisys processing
4. Channel mapping
5. SW Mixer

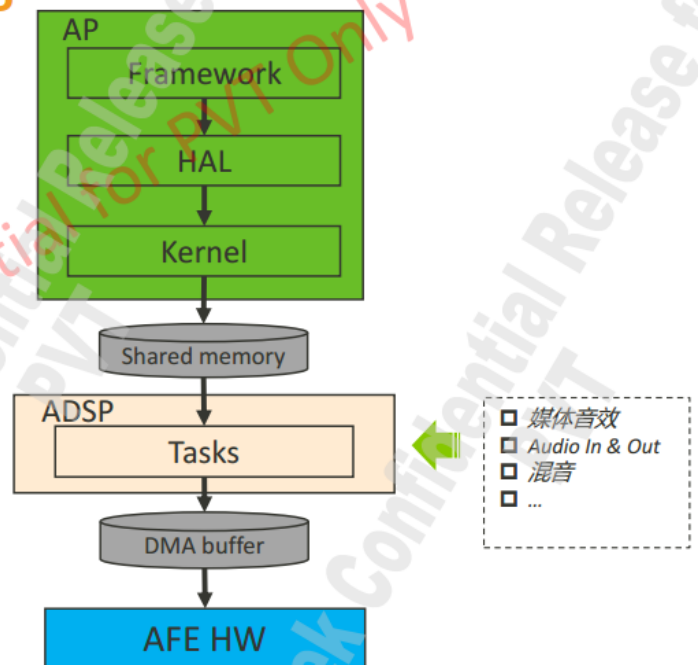


8676 vs 8668, Adsp的功能没有变化

Go through ADSP and bypass ADSP



Audio playback and capture flow w/o ADSP



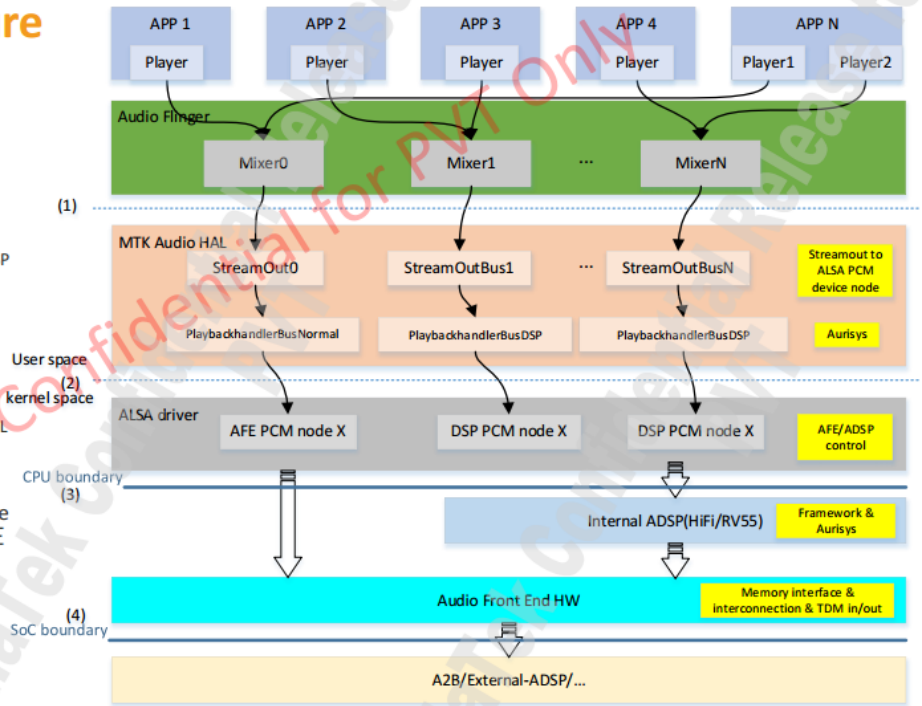
Audio playback flow w/ ADSP

- ☐ 媒体音效
- ☐ Audio In & Out
- ☐ 混音
- ☐ ...

8676 vs 8668, 走adsp的和bypass adsp的软件架构没有变化

AAOS Audio Architecture

- AudioFlinger -> (HIDL/AIDL) -> MTK Audio HAL
- MTK Audio HAL -> (tinyALSA) -> ALSA driver
- Case 1: go through INT-ADSP
 - ALSA driver -> (MTK IPI & share DRAM) -> INT-ADSP
 - INT-ADSP -> (DRAM) -> Audio Front End HW -> DL MEMIF -> InterConn -> TDM out -> external DSP/A2B
- Case 2: bypass INT-ADSP
 - ALSA driver -> (DRAM) -> Audio Front End HW -> DL MEMIF -> InterConn -> TDM out -> external DSP/A2B
- 在Audio HAL将多路数据通过DSP PCM device node或者AFE PCM device node写到对应通路，最终在AFE HW 可以做 HW Mix再输出



8676 vs 8668, aaos软件架构没有变化，走adsp的走playbackhandlerbusdsp，bypass adsp的走playbackhandlerbusnormal。